



VENDOR iQ

VENDOR MANAGEMENT OPTIMISED

Business Conclusion Report

VendorIQ – Inventory & Vendor Analytics

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Executive Summary

VendorIQ is an end-to-end analytics initiative built on the company's vendor, purchasing, sales, and inventory data. The project combines SQL-based business analysis, Python-driven exploratory data analysis and statistical testing, and an interactive Power BI dashboard suite to help management understand vendor performance, procurement efficiency, inventory health, and overall profitability. The findings below summarize the headline metrics captured across the Executive Overview, Vendor & Procurement, and Inventory & Product dashboards.

Total Revenue	\$452.1M
Total Purchase Cost	\$321.90M
Ending Inventory Value	\$79.7M
Active Vendors	129
Distinct Products	268K
Average Selling Price	\$14
Inventory Turnover	3.75x
Total Purchase Orders	6K

Overall, the business generated approximately \$130M in gross profit ($\approx 28.8\%$ blended margin) across the analysis period. Revenue and profit are heavily concentrated among a small set of strategic vendors, procurement costs benefit meaningfully from bulk ordering, and inventory levels are generally well aligned with sales demand — though a subset of premium products ties up a disproportionate share of working capital. Full statistical evidence for each of these conclusions is presented in the following sections.

Objectives

The analysis was designed around five core business objectives:

- Evaluate Vendor Performance – identify which vendors drive the majority of revenue and profit, and which underperform.
- Optimize Purchasing Strategy – determine whether purchase volume and vendor selection influence unit procurement cost.
- Improve Inventory Management – assess whether inventory levels are aligned with demand and identify capital tied up in slow-moving or high-value stock.
- Improve Profitability – quantify how vendor quality relates to profit margin, and establish reliable benchmarks for margin and price.
- Support Data-Driven Decision Making – provide management with statistically validated, dashboard-ready insights for procurement, inventory, and vendor management decisions.

Dataset Description

The dataset represents a full year of vendor, purchasing, sales, and inventory operations for a multi-store liquor retail business, stored in a SQL Server database named Inventory. Six core tables were used throughout the analysis:

Table	Description	Rows	Columns
begin_inventory	Opening stock position by store/brand	206,529	9
end_inventory	Closing stock position by store/brand	224,489	9
purchases	Line-level purchase order transactions	2,372,474	16
purchase_prices	Vendor list price / cost reference table	12,260	9
sales	Line-level point-of-sale transactions	12,825,363	14
vendor_invoice	Vendor invoice, freight & approval records	5,543	10

These transactional tables were modeled into a star schema — DimDate, DimVendor, DimStore, and DimProduct dimension tables surrounding a Measures table — to support the Power BI dashboards. The relationship model is shown below.

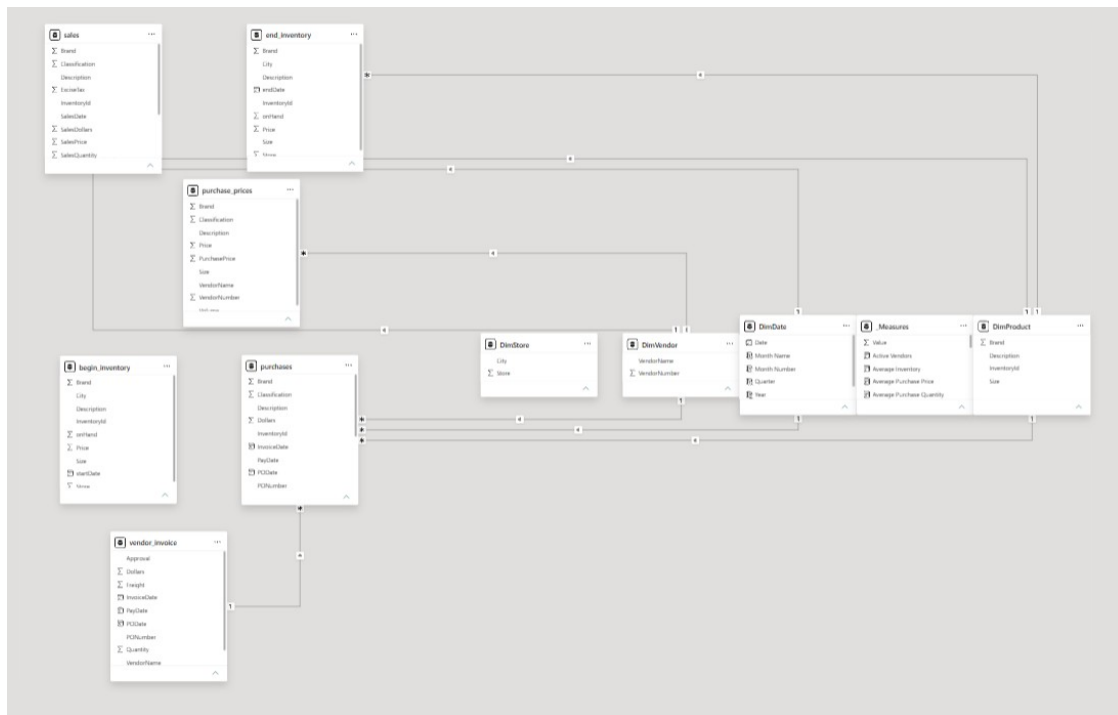


Figure 1 — VendorIQ Data Model (Star Schema: Sales, Purchases, Inventory & Vendor Dimensions)

SQL Analysis

A dedicated T-SQL script (Business_Solution.sql) was written against the Inventory database to answer 48 business questions spanning vendor performance, brand performance, purchasing behavior, inventory health, and profitability trends. The queries make extensive use of joins, aggregate functions, and window functions.

Key Query Categories

- Vendor performance — total sales, purchase value, profit, and profit margin by vendor, including top/bottom-10 rankings using RANK() and DENSE_RANK().
- Brand performance — revenue, profit, profit margin, and revenue contribution percentage by brand.
- Purchasing metrics — average purchase quantity and price, and vendors offering the highest/lowest average purchase prices.
- Inventory analytics — current inventory quantity and value by brand/vendor, inventory turnover ratio, slow-moving and dead-stock identification, and capital locked in inventory.
- Time-series trends — monthly sales, profit, and purchase trends, cumulative (running) revenue and profit using window aggregates, and month-over-month growth using LAG().

Representative Query — Vendor Profitability

```
SELECT s.VendorNo, s.VendorName,  
       SUM(s.SalesDollars) AS TotalSales,  
       SUM(s.SalesQuantity * p.PurchasePrice) AS TotalCost,  
       ROUND(SUM(s.SalesDollars - (s.SalesQuantity * p.PurchasePrice)),2) AS  
TotalProfit  
FROM sales s  
JOIN purchase_prices p  
  ON s.Brand = p.Brand AND s.Description = p.Description AND s.Size = p.Size  
GROUP BY s.VendorNo, s.VendorName;
```

This query forms the backbone of the profitability analysis used throughout the statistical notebook and dashboards, joining transactional sales records to vendor purchase prices to derive per-vendor profit and margin.

Python EDA

Exploratory data analysis and data cleaning were performed in Python (pandas) prior to loading the cleaned tables into SQL Server via SQLAlchemy. The following data quality steps were applied:

- Missing values: end_inventory.City (1,284 nulls) filled with "Unknown"; a single incomplete purchase_prices row (missing description, size, and price) was dropped.
- Reference-data repair: 3 purchases records with a missing Size were corrected by cross-referencing the brand's known size in purchase_prices.
- Approval status: 5,169 vendor_invoice records with a blank Approval field were labeled "Pending" rather than left null.
- Duplicates: all six tables were checked and confirmed to contain zero duplicate rows.
- Standardization: leading/trailing whitespace stripped from all text columns and column headers; all date-like columns coerced to proper datetime types.
- Validation: all numeric columns were checked for negative values (none found), confirming clean, non-erroneous transactional data.

- The cleaned tables were then uploaded to the Inventory SQL Server database for use by the SQL analysis and Power BI dashboards.

Statistical Analysis

A deeper statistical analysis was performed in Python (SciPy, pandas, matplotlib/seaborn) covering correlation analysis, distribution analysis, outlier detection, hypothesis testing, and confidence interval estimation.

Correlation Analysis

Relationship	Correlation (r)	p-value	Interpretation
Purchase Quantity vs. Unit Cost	-0.072	0.422	No meaningful relationship
Sales vs. Profit	0.990	<0.001	Very strong positive
Purchase Price vs. Sales Price	0.998	<0.001	Near-perfect positive
Inventory On-Hand vs. Sales Qty	0.844	<0.001	Strong positive

Vendors that generate more sales generate proportionally more profit ($r = 0.990$), and selling prices track procurement costs almost perfectly ($r = 0.998$), confirming that the company's pricing strategy consistently reflects supplier cost changes. Inventory levels are strongly, though not perfectly, aligned with demand ($r = 0.844$). Notably, purchase volume alone shows no significant relationship with unit cost at the per-transaction level ($r = -0.072$, $p = 0.422$) — cost reduction instead comes from bulk-order thresholds, confirmed later by hypothesis testing.

Distribution Analysis

Business Metric	Distribution	Skewness	Insight
Revenue	Highly right-skewed	4.439	Concentrated among a few vendors
Profit	Highly right-skewed	3.844	Depends on a limited vendor group
Profit Margin	Moderately left-skewed	-0.749	Mostly consistent (30-35%)
Inventory Value	Extremely right-skewed	10.046	Capital concentrated in few SKUs

Outlier & Hypothesis Testing Highlights

- 901 products (9.33%) were flagged as high inventory-value outliers, representing \$44.85M in inventory investment — led by Jack Daniel's No. 7 Black, Ketel One Vodka, Johnnie Walker Black Label, Absolut 80 Proof, and Tito's Handmade Vodka.
- Only 4 vendors (3.20% of the supplier base) were low-margin outliers, but they account for just 0.09% of total revenue — a negligible financial risk.
- 15 vendors (12% of the supplier base) were classified as strategic high performers, contributing 37.73% of total revenue; 7 underperforming vendors contributed only 0.02% of revenue.
- Mann-Whitney U test: bulk purchases (≥ 10 units) average \$9.85/unit vs. \$14.62/unit for smaller orders — a statistically significant 32.6% cost reduction ($p < 0.001$).
- Mann-Whitney U test: high-performing vendors average a 36.88% profit margin vs. 29.63% for other vendors — a statistically significant ~24.5% margin improvement ($p < 0.001$).

Confidence Intervals (95%)

- Average vendor profit margin: 30.55%, CI [29.43%, 31.67%] — a precise, reliable profitability benchmark.
- Average purchase price: \$12.05, CI [\$12.03, \$12.07], based on 2.37M+ purchase transactions — an extremely stable procurement benchmark.

Dashboard 1 Summary — Executive Overview

The Executive Overview dashboard provides a top-level view of company performance for 2024: \$452.1M in total revenue, \$79.7M in ending inventory value, 129 active vendors, 268K distinct products, a \$14 average selling price, and an inventory turnover of 3.75x. Revenue trended downward across the year from a December-2023 high, and the top vendors — Diageo North America, Martignetti Companies, and Pernod Ricard USA — together account for the largest share of revenue, with Jack Daniel's No. 7 Black leading product-level revenue.

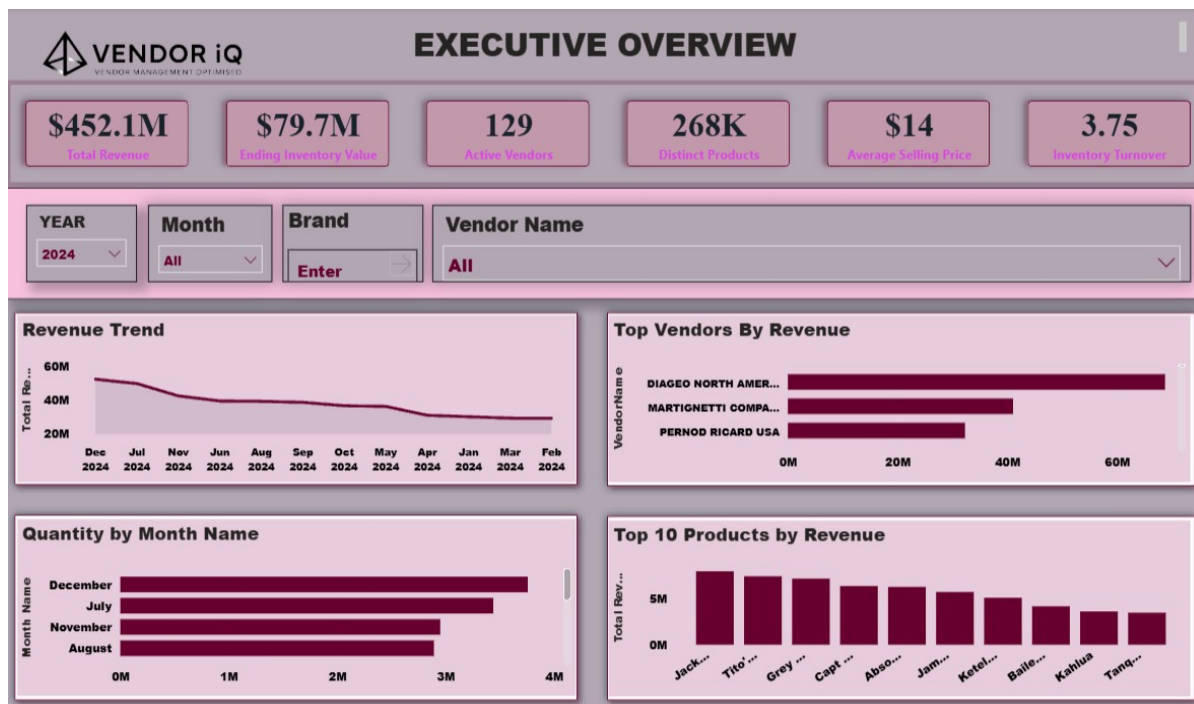


Figure 2 — Executive Overview Dashboard

Dashboard 2 Summary — Vendor & Procurement Analytics

The Vendor & Procurement Analytics dashboard totals \$321.90M in purchase cost across roughly 34M units and 6K purchase orders, at an average purchase price of \$9.58 and average purchase quantity of 14.16 units. Purchase cost rose sharply between April and August before leveling off. Vendor spend is highly concentrated: Diageo North America (\$51M), Martignetti Companies (\$28M), Jim Beam Brands (\$24M), and Pernod Ricard (\$24M) are the largest sources of procurement spend. The embedded Mann-Whitney U test ($p < 0.001$) confirms bulk purchases carry significantly lower unit prices, while average purchase price stays stable across the year — evidence of consistent, well-managed procurement costs.

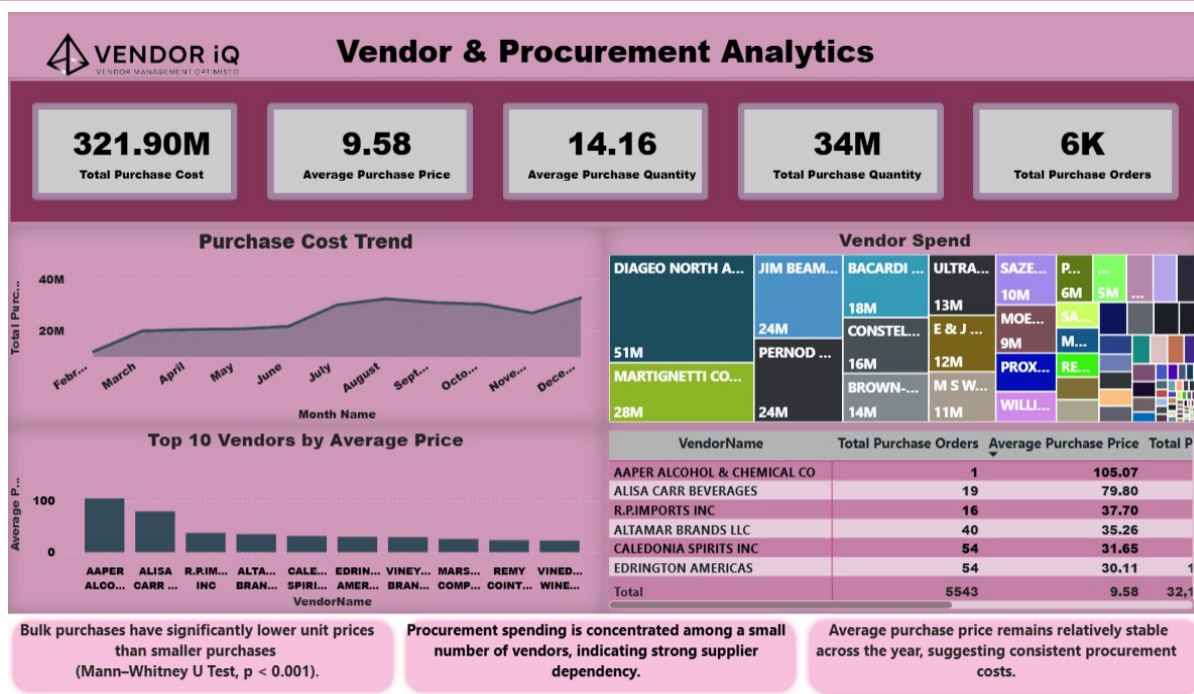


Figure 3 — Vendor & Procurement Analytics Dashboard

Dashboard 3 Summary — Inventory & Product Analytics

The Inventory & Product Analytics dashboard shows inventory value growing from \$68.05M to \$79.70M (an \$11.65M increase) while unit count grew from 4M to 5M. Capt. Morgan Spiced Rum, Jack Daniel's No. 7 Black, and Absolut 80 Proof hold the largest ending inventory value, while Smirnoff and Tito's lead in sales quantity. Revenue share is concentrated among the top 5 vendors, led by Diageo North America (35%) and Martignetti Companies (21%), reinforcing the vendor-concentration pattern seen throughout the statistical analysis.

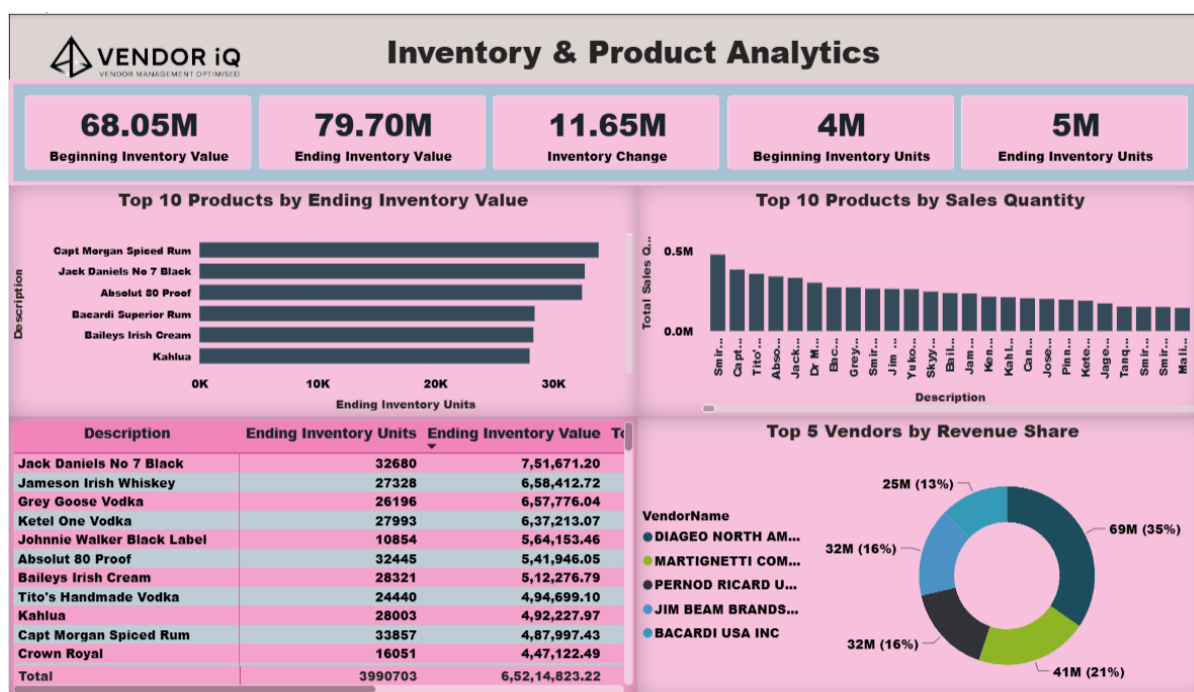


Figure 4 — Inventory & Product Analytics Dashboard

Key Business Insights

- Revenue and profit are concentrated among a small group of vendors: just 15 vendors (12% of the supplier base) drive 37.73% of total revenue, while 7 underperforming vendors contribute only 0.02%.
- Sales and profit move almost perfectly together ($r = 0.990$) — profitability is primarily a function of driving revenue through the strongest vendors, not margin variation.
- Selling prices closely track procurement costs ($r = 0.998$), showing that the company's pricing strategy consistently protects margin as supplier costs change.
- Bulk purchasing materially reduces unit cost (32.6% lower, $p < 0.001$) — but purchase volume alone shows no significant correlation with price at the transaction level, meaning vendor negotiation and contract terms matter more than raw order size.
- High-performing vendors sustain meaningfully higher profit margins (36.88% vs. 29.63%, $p < 0.001$), confirming that vendor selection has a measurable, statistically validated impact on profitability.
- Inventory is generally well aligned with demand ($r = 0.844$), but 901 products (9.33%) are high-value inventory outliers tying up \$44.85M in working capital, led by premium spirits brands.
- Company-wide benchmarks are highly reliable: average profit margin of 30.55% (95% CI: 29.43%–31.67%) and average purchase price of \$12.05 (95% CI: \$12.03–\$12.07), both estimated from very large, statistically robust samples.

Business Recommendations

Vendor Management

- Prioritize and strengthen relationships with the 15 strategic vendors driving 37.73% of revenue through preferential terms, joint planning, and dedicated account management.
- Address the small number of low-margin outlier vendors (4 vendors, 0.09% of revenue) individually through targeted pricing reviews rather than organization-wide changes.
- Monitor vendor concentration risk — diversify the supplier base where feasible to reduce dependency on a handful of top-revenue vendors.

Procurement Strategy

- Continue and expand bulk-order consolidation where demand is stable, given the statistically confirmed 32.6% unit-cost reduction from bulk purchasing.
- Focus procurement negotiations on vendor-specific contract terms and pricing agreements rather than assuming volume alone drives savings.

Inventory Management

- Apply closer demand forecasting and review cycles to the 901 high-value inventory outlier products (representing \$44.85M in capital) to reduce carrying cost and stockout/overstock risk.
- Use the strong inventory–sales correlation ($r = 0.844$) as a baseline for replenishment planning, while investigating SKUs where inventory levels diverge from actual demand.

Pricing & Profitability

- Maintain periodic pricing reviews to keep the near-perfect purchase-to-sales price alignment ($r = 0.998$) intact as supplier costs shift.
- Use the 30.55% average margin (95% CI: 29.43%–31.67%) as the benchmark for evaluating vendor and product profitability going forward.

Future Improvements

- Extend the analysis to a multi-year time series to evaluate seasonality, year-over-year growth, and the sustainability of current vendor-concentration patterns.
- Incorporate freight, invoice-approval delay, and payment-term data from vendor_invoice into a full landed-cost and vendor-reliability scorecard.
- Build a predictive model (e.g., regression or time-series forecasting) for demand and inventory optimization, rather than relying solely on descriptive/statistical analysis.
- Automate the Power BI dashboards with a scheduled refresh against the live SQL Server database, and add drill-through pages for individual vendors and products.
- Layer in external factors (seasonal promotions, holidays, regional demand differences) to explain the variance not captured by inventory-sales correlation alone.

Conclusion

The VendorIQ analysis demonstrates that the business operates a financially healthy but vendor-concentrated model: a relatively small group of high-performing suppliers generates the large majority of revenue and profit, procurement costs are effectively managed through bulk purchasing, and pricing strategy reliably protects

margin against cost changes. Inventory planning is broadly aligned with sales demand, though a defined subset of premium products warrants closer working-capital management. Every conclusion in this report is backed by statistically validated evidence — correlation analysis, distribution and outlier detection, hypothesis testing, and confidence intervals — giving management a reliable, data-driven foundation for vendor management, procurement, inventory, and pricing decisions going forward.

— End of Report —